

Project Proposal (Proposed Solution) Report

Date	14 July 2026
Team ID	
Project Name	HDI Predictor (Human Development Index Predictor)
Team Size	1 (Individual Project)
Maximum Marks	5 Marks

Project Overview

Field	Details
Objective	Predict a country's Human Development Index and development tier from four measurable indicators, through an accessible, interactive web app.
Scope	Covers data collection through a live, publicly deployed 3-dashboard application (Home, Predict, History) backed by a trained regression model.

Problem Statement

Field	Details
Description	HDI is meaningful but not directly computable by most users; existing resources are static tables, not interactive tools.
Impact	Without an interactive tool, policymakers and researchers spend extra time on manual computation and lose the ability to quickly test scenarios.

Proposed Solution

Field	Details
Approach	Train a regression model on real UNDP 2021 data (log-transforming GNI to mirror HDI's true formula), then serve it through a Flask web app.
Key Features	• HDI + tier prediction from 4 indicators • Manual, blank-by-default input form (no misleading auto-fill) • Optional real-country autofill for reference • Persistent prediction history with filtering, delete, and trend chart • Home dashboard with global stats and EDA visuals

Resource Requirements

Resource Type	Description	Specification / Allocation
Computing Resources	Local training + inference	Standard CPU; no GPU required
Hosting	Public deployment	Render free-tier web service (Gunicorn)
Dataset	Training data	UNDP Human Development Report 2021 (195 countries)